

Appendices

Appendix A: Details for Data Cleaning

In ProQuest TDM Studio, the texts for The Wall Street Journal prior to 1984 and for The Washington Post prior to 1987 were produced by using Optical Character Recognition (OCR) technology. To ensure high-quality data, I tokenized these articles and conducted an OCR quality check by calculating the percentage of words in each article that match entries in a comprehensive dictionary of existing English words derived from the HathiTrust corpus created by Underwood and Levin (accessed through <https://github.com/tedunderwood/DataMunging/blob/master/rulesets/MainDictionary.txt>). I included only texts with an OCR accuracy rate above 80%. I also filtered out non-substantial content, such as advertisements, legal notices, and stock quotes from the dataset.

As the focus of this paper is on the United States, I used two methods to screen out non-U.S. articles while preserving those that might be related to U.S. foreign relations. First, for articles with location tags (N=3,110,058), I created a list of non-U.S. country and city names and removed articles that had only non-U.S. location tags. For example, an article tagged with only “Japan” would be removed, but one tagged with “Japan; U.S.” would be retained. This step removed 844,447 articles. Second, for articles without location tags (N=6,785,510), I removed those that met two criteria: (1) the article’s title contained non-U.S. locations, and (2) the article’s title did not mention U.S. locations (e.g., U.S., U.S.A., American, or the names of the 50 states). This step further removed 400,765 articles.

Appendix B: Procedure for Selecting Bigrams

I sampled 50,000 articles from the dataset and used the `textstat_collocations` function in the `quanteda.textstats` package to retain bigrams. There are two metrics, λ and z , in the `textstat_collocations` function. The metric λ measures the strength of association between words using a log-likelihood ratio test, assessing how likely the words form a collocation compared to random co-occurrence. The z score represents the statistical significance of the observed collocation frequency compared to its expected frequency under independence. Based on reviewing selected bigrams, I chose the threshold of $\lambda \geq 5$ & $z \geq 50$ to ensure that only bigrams with a high level of distinctiveness and are significantly over-represented are included. For example, the bigrams “food stamps” ($\lambda = 8.47$, $z = 74.12$) and “child care” ($\lambda = 5.79$, $z = 104.15$) are retained. These bigrams are necessary to retain because they convey distinct meanings that differ from the individual words (e.g., “food stamps” is distinct from “food” and “stamps” separately). I only included bigrams that appear at least 25 times in the sample for analysis. In total, 2,342 bigrams are retained.

In addition, I examined collocations with the terms “liberal” and “conservative” in them. I retained the bigrams (and some trigrams) that are typically used in non-political senses to make sure that they are distinct from “liberal” and “conservative” when embedding vectors are calculated. This includes the following terms:

liberal arts, neo liberalism, liberal interpretation, liberal definition, liberal use, liberal lending, liberal application, liberal dress code, liberal return policy, liberal return policies, liberal pricing, liberal estimate, liberal estimates, liberal portion, liberal dose, liberal amount, liberal amounts, liberal heap, liberal splash, liberal substitution, liberal scoring, conservative investment, conservative investments, conservative investor, conservative investors, conservative management, conservative accounting, conservative lending, conservative portfolio, conservative portfolios, conservative estimate, conservative estimates, conservative projection, conservative projections, conservative forecast, conservative forecasts, conservative forecaster, conservative forecasters, conservative amount, conservative amounts, conservative bids, conservative assumptions, conservative assumption, conservative definition, conservative color, conservative colors, conservative look, conservative outfit, conservative wardrobe, conservative clothing, conservative styling, conservative suit, conservative suits, conservative dress, conservative attire, conservative mode, conservative painting, conservative paintings, conservative sofas, conservative contemporary, conservative musical, conservative harmonic, conservative repertory, conservative repertoire, conservative idiom, conservative production, conservative productions, conservative play, conservative play-calling, conservative coach, conservative coaching, conservative passing, conservative shot, conservative

shots, conservative game, conservative composer, conservative orchestra, conservative houses, aesthetically conservative, musically conservative, artistically conservative, financially conservative, stylistically conservative, harmonic conservatism

I also retained common proper nouns that include the terms “liberal” and “conservative,” such as “liberal democratic party,” “progressive conservative party,” “liberal front party,” “british conservative,” “britain conservative,” and “conservative party.” I retained these terms because the parties referenced are non-U.S. (e.g., the Conservative Party in the U.K. and the Liberal Democratic Party in Japan), and I want to ensure they don’t interfere with the “liberal” and “conservative” embedding vectors.

Appendix C: Selecting The Optimal Number of Topics

To select the optimal number of topics for a topic model, I used the `ldatuning` package in R. The process began by creating a document-term matrix from a sample of 50,000 articles. Using the `FindTopicsNumber` function, I specified a range of potential numbers of topics (200, 250, 300, 350, 400, and 500) and applied the Gibbs sampling method to fit Latent Dirichlet Allocation (LDA) models. The CaoJuan 2009 metric measures the average cosine similarity between topics, where a lower value indicates better separation between topics, while the Deveaud 2014 metric maximizes the distance between topics, with a higher score indicating a better model fit. After running the models, I visualized the results using the `FindTopicsNumber_plot` function, identifying the optimal number of topics by locating the minimum point for the CaoJuan metric and the maximum point for the Deveaud metric (see Figure C1). The results indicate that the 350-topic solution minimizes the CaoJuan 2009 metric. Meanwhile, the Deveaud 2014 metric shows diminishing improvement beyond 350 topics, suggesting that the analytical gains from decreasing the number of topics become more limited after this point. Therefore, 350 topics represent the optimal choice for effectively separating the topics.

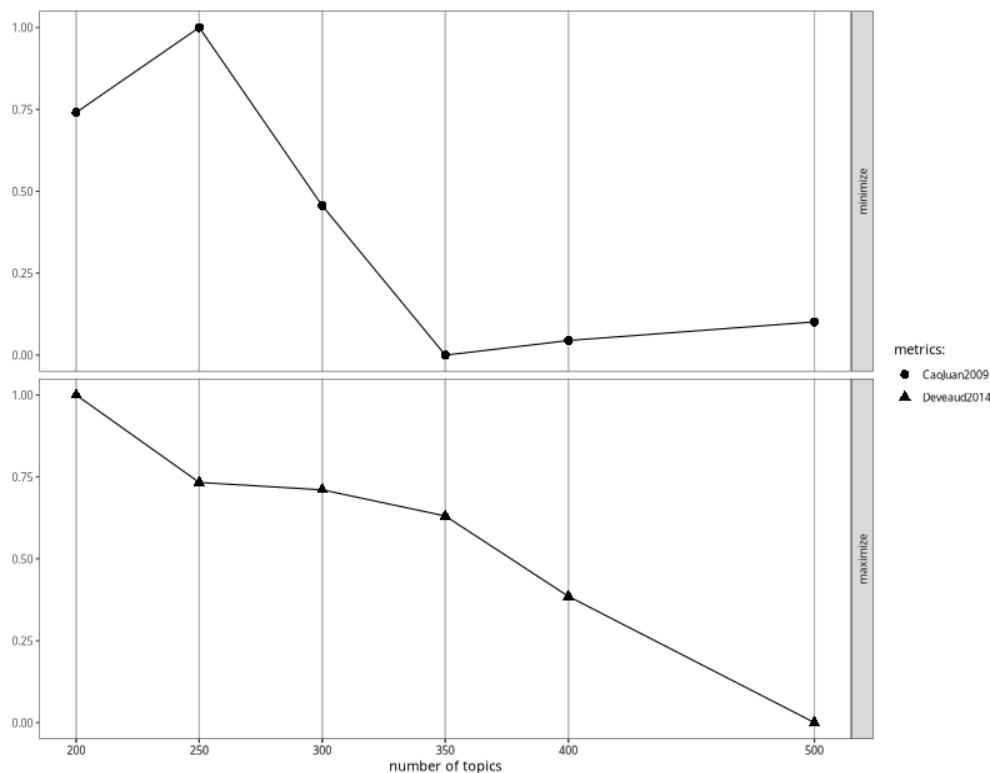


Figure C1. Optimal Number of Topics for LDA Model

Appendix D: Topic-Model-Based Dictionaries

This appendix lists the 25 issue domains in alphabetical order, the 131 topics assigned to them, and the terms included in each topic dictionary. The Movie and Baseball dictionaries each combine two closely related topics identified by the topic model.

Animals

Animal: dog, animals, animal, dogs, bird, cat, birds, cats, pet, deer, zoo, pets, animal-control, owl, bats, pigs, humane-society, bat, tail, owls, rabies, animal-shelter, rats, fur, creatures, veterinarian, monkeys, feral, geese, panda, pit-bull, skunk, kennel, turkeys, veterinary, puppy, rooster, canine, squirrel, raccoon, coyote, rabbits

Marine Biology: fish, fishing, waters, marine, salmon, species, fishermen, shark, crabs, aquarium, trout, oyster, oysters, sharks, seals, chesapeake-bay, coral, fisheries, perch, whale, reef, crab, bait, anglers, bluefish, tuna, algae, rockfish, flounder, turtles, lobsters, dolphin, striped-bass, spawning, catfish, oceanic, clams, octopus, goldfish, shellfish, shad, squid, swordfish

Wildlife: forest, alaska, conservation, lands, wildlife, forests, environmentalists, habitat, forest-service, wilderness, timber, wolves, arctic, wetlands, endangered-species, audubon, logging, environmental-groups, grouse, ecology, conservationists, grazing, ecological, wildlife-refuge, ranchers, alaskan, habitats, ecosystems

Creative Arts

Art: artist, painting, paintings, painted, portrait, sculpture, drawings, painter, abstract, photography, portraits, prints, picasso, canvas, retrospective, stella, landscapes, sculptor, gogh, matisse, photographic, calder, abstraction, painters, canvases, collage, warhol, degas, andy-warhol, collages, impressionist, vermeer, rothko, geometric, self-portrait, louvre, expressionist, monet

Broadway: theater, broadway, musical, actors, shakespeare, productions, playwright, theatrical, costumes, onstage, staging, artistic-director, musicals, costume, previews, playwrights, troupe, mamet, sondheim, brantley, cabaret, hammerstein, macbeth, tony-award, neil-simon, actresses, folger, rehearsals, monologues

Classical Music: dance, opera, ballet, orchestra, dancers, concert, symphony, piano, composer, pianist, concerts, mozart, philharmonic, ensemble, conductor, bach, beethoven, composers, lincoln-center, soprano, classical-music, violinist, sonata, violin, recital, quartet, choreography, symphony-orchestra, repertory, carnegie-hall, concerto, tenor, danced, operas, cello, balanchine, metropolitan-opera, orchestras, tchaikovsky, orchestral, brahms, ballet-theater, schumann, choral

Fashion: fashion, wear, designer, dress, shoes, designers, pants, leather, dresses, jeans, shirt, jacket, shirts, fabric, boots, jackets, sneakers, silk, coat, coats, runway, skirts, skirt, accessories, outfit, fabrics, socks, sweater, prada, wool, wardrobe, chic, gucci, sweaters, footwear, ralph-lauren,

trousers, lace, denim, garment, vuitton, chanel, creative-director, dior, sleeves, sportswear, calvin-klein

Literature: book, books, novel, writers, literary, authors, publishers, poetry, fiction, literature, novels, prose, memoir, poem, poems, novelist, bookstore, knopf, random-house, manuscript, poets, essays, hemingway, reviewer, bookstores, booksellers, doubleday, didion, novelists, manuscripts, bestseller

Movie: disney, entertainment, hollywood, studios, sony, paramount, streaming, box-office, blockbuster, marvel, warner-bros, walt-disney, dvd, animation, walt-disney-co, motion-picture, star-wars, paramount-pictures, pixar, spider-man, film, movie, films, movies, documentary, actor, actress, starring, script, cinema, filmmakers, filmmaker, screenplay, film-festival, filming, thriller, sundance, cinematic, documentaries, sequel, screenwriter, scorsese, subtitles, academy-award, sci-fi, cannes, filmmaking, keaton, niro, costner

Museum: museum, art, artists, exhibition, gallery, exhibit, museums, galleries, smithsonian, curator, artifacts, exhibitions, exhibits, collections, collectors, metropolitan-museum, sculptures, art-museum, contemporary-art, national-gallery, artwork, modern-art, ceramics, fine-arts, curators, national-endowment, moma, hirshhorn, smithsonian-institution, sculpture-garden, art-gallery, artworks, nea, permanent-collection, watercolors, mixed-media

Music: band, album, songs, song, jazz, guitar, musicians, blues, albums, lyrics, sing, hip-hop, solo, bands, cd, guitarist, rap, sings, drums, rapper, songwriter, beatles, saxophonist, vocals, bluegrass, guitars, ballad, melody, grammy, melodies, pop-music, rock-n-roll, saxophone, fender, acoustic, singer-songwriter, ballads, harmonies, vocalist, reggae

Crime and Policing

Burglary: stolen, residence, robbed, burglary, break-in, robbers

Criminal: trial, prosecutors, jury, convicted, indictment, testimony, testified, jurors, guilty, defendants, prosecution, pleaded-guilty, defendant, witnesses, conspiracy, sentenced, verdict, courtroom, prosecutor, indicted, testify, grand-jury, sentencing, district-attorney, pleaded, felony, federal-prosecutors, acquitted, defense-lawyers, juror, racketeering, found-guilty, conspiring, defense-attorneys, indictments, juries, bribery, plead-guilty, prosecuting, district-attorney-office, guilty-plea, extortion, perjury, cross-examination, prison-sentence, bribe, witness-stand, wire-fraud

CriminalJustice: prison, jail, prisoners, inmates, rape, sentence, punishment, prisons, death-penalty, detention, execution, sentences, inmate, parole, prisoner, offenders, guards, detainees, torture, criminal-justice, sexual-assault, correctional, criminals, offenses, probation, detained, raped, jails, imprisoned, incarcerated, murders, executions, rikers, pardon, imprisonment, incarceration, offender, prosecuted, clemency, behind-bars, prosecutions, violent-crime, felons, raping

Homicide: incident, medical-examiner, stabbed, slaying, autopsy, fatally-shot, stabbing, second-

degree-murder, coroner, first-degree-murder, assailant, attempted-murder, slayings, gunshot-wound

Investigation: allegations, investigators, fbi, investigations, inquiry, probe, f.b.i, investigated, federal-bureau, misconduct, criminal-charges, affidavit, criminal-investigation, allegation, laundering, bribes, fbi-agents, informant, independent-counsel, special-prosecutor, criminal-complaint

Police: police, officers, police-officers, police-department, police-officer, detective, patrol, policing, detectives, d.c-police, cops, nypd, police-departments, homicide, troopers, precinct, internal-affairs, police-commissioner, uniformed, racial-profiling, homicides, law-enforcement-agencies, dispatcher

Economic Issues

Agriculture: farm, farmers, corn, agriculture, farms, farmer, agricultural, crop, meat, milk, crops, dairy, grain, wheat, beef, usda, farming, cow, cattle, cows, agriculture-department, growers, harvest, chickens, genetically-modified, poultry, perdue, monsanto, hog, hogs, livestock, deere, acreage, bushels, oats, calves, soy, manure

Business: business, entrepreneurs, small-businesses, entrepreneurship, small-business, entrepreneurial, sba, small-business-owners

Economics: economists, commerce-department, annual-rate, consumer-spending, retail-sales, chief-economist, unemployment-rate, consumer-confidence, price-index, durable-goods, gross-domestic-product, personal-income, payrolls, housing-starts, year-over-year, jobless-rate, cpi, jobless-claims, gnp

Economy: recession, unemployment, economic-growth, stimulus, gdp, globalization, economic-recovery, stimulus-package, economic-advisers, recessions

Fiscal: budget, cuts, fiscal, deficit, tax-cuts, deficits, surplus, tax-increases, spending-cuts, budget-deficit, budget-cuts, debt-ceiling, deficit-reduction, outlays, congressional-budget, debt-limit, cbo, balanced-budget, raise-taxes, budgetary, surpluses, raising-taxes

Labor: union, labor, strike, unions, bargaining, pension, teamsters, uaw, concessions, afl-cio, labor-relations, unionized, picket, nonunion, collective-bargaining, nlr, organized-labor, walkout, auto-workers, machinists, wage-increases, strikers, steelworkers, cost-of-living

Monetary Policy: fed, inflation, federal-reserve, yields, central-bank, interest-rate, monetary-policy, greenspan, bond-market, money-supply, central-banks, bond-prices, bond-yields, short-term-interest-rates, federal-reserve-bank, inflation-rate, borrowing-costs, treasury-bonds, inflationary, 10-year-treasury, federal-funds, inflationary-pressures

Taxation: tax, taxes, irs, capital-gains, income-tax, deduction, ira, deductions, internal-revenue, income-taxes, taxation, taxed, tax-code, tax-breaks, deduct, tax-reform, tax-returns, iras, i.r.s, deductible, taxing, tax-deferred, adjusted-gross-income, depreciation, taxable-income, brackets, filers, roth-ira, excise

Trade: steel, imports, tariffs, exports, export, import, aluminum, imported, tariff, trade-deficit,

quotas, trade-representative, nafta, steelmaker, exporters, steelmakers, supply-chains, free-trade, raw-materials, exporting, protectionism, protectionist, gatt, wto, alcoa, exporter

Welfare & Poverty: homeless, welfare, poverty, disabled, shelter, disability, disabilities, shelters, homelessness, neediest-cases, food-stamps, handicapped, needy, mentally-ill, legal-aid, catholic-charities, homeless-families, welfare-reform, social-worker, welfare-recipients, safety-net, wheelchairs

Workers: workers, employees, jobs, employers, employee, hiring, employer, wages, workplace, layoffs, minimum-wage, workforce, unemployed, labor-department, human-resources, labor-force, sick-leave, furloughs, low-wage, laid-off

Education

Education: school, students, schools, teachers, teacher, public-schools, math, classroom, elementary, middle-school, grades, classrooms, educators, kindergarten, school-districts, instruction, charter-schools, curriculum, principals, test-scores, charter-school, elementary-schools, mathematics, classmates, graders, after-school, class-sizes, physical-education, pupils, preschool, vouchers, junior-high, tutoring, superintendents, pta

Higher Education: college, student, campus, colleges, universities, graduate, academic, tuition, faculty, graduates, admissions, higher-education, professors, alumni, scholarships, campuses, graduation, semester, financial-aid, undergraduate, graduating, faculty-members, undergraduates, liberal-arts, ivy-league, freshmen, commencement, uc, cuny, vocational, regents, provost, hazing

Environment

Air Pollution: tobacco, epa, smoking, chemical, cigarette, toxic, asbestos, smoke, cigarettes, cleanup, environmental-protection-agency, hazardous, contamination, ozone, e.p.a, tobacco-companies, air-quality, pesticides, contaminated, hazards, superfund, pesticide, pollutants, wastes, air-pollution, lung-cancer, sludge, soot

ClimateChange: climate, climate-change, emissions, solar, carbon, global-warming, pollution, renewable-energy, carbon-dioxide, fossil-fuels, clean-energy, power-plants, clean-air, warming, recycling, sustainability, fossil-fuel, solar-panels, methane, greenhouse-gases, esg, renewable, carbon-emissions, greenhouse-gas, greenhouse-gas-emissions, hydropower, coal-fired, greenhouse, gases, emit

Disaster: storm, hurricane, earthquake, storms, winds, flooding, hurricanes, flooded, fema, emergency-management, debris, quake, evacuation, tornado, floods, tropical-storm, tornadoes, earthquakes, rescuers, evacuees, noaa, death-toll

Weather: cloudy, degrees, rain, temperature, sunny, temperatures, showers, weather-service, partly-cloudy, thunderstorms, humid, humidity, partly-sunny, cooler, heat-wave, fahrenheit, precipitation, windy, meteorologist, temp

Wildfires: fire, fires, firefighters, burning, fire-department, flames, wildfire, blaze, wildfires, firefighter, evacuated, explosions, fatalities, evacuate

Financial Market

Bankruptcy: bankruptcy, bankruptcy-court, reorganization, bankruptcy-protection, bankruptcy-code, reorganization-plan, bankruptcies, bankruptcy-filing, bankruptcy-law, debtors, debtor, bankruptcy-proceedings, defaulted, involuntary

Banks: bank, banks, banking, bankers, deposits, fdic, thrift, citigroup, wells-fargo, financial-institutions, jpmorgan, citicorp, j.p-morgan, thrifts, depositors, federal-deposit-insurance, bancorp, deutsche-bank, bankamerica, banking-system, citibank, pnc, hsbc, wachovia, chase-manhattan, home-loan, citi, bank-holding-company, deposit-insurance, citigroup-inc, bank-holding, savings-loan-association, nonperforming

Commodity Market: futures, commodity, inventories, new-york-mercantile-exchange, soybean, cocoa, speculators, futures-prices, soybeans, commodity-markets, price-increases, chicago-mercantile-exchange, bearish, futures-contracts

Corporate: shareholders, merger, stake, takeover, shareholder, tender-offer, buyout, icaohn, buffett, takeovers, stockholders, takeover-bid, berkshire, leveraged-buyout, berkshire-hathaway, buy-out, revlon, proxy-fight, suitor, recapitalization, hostile-takeover, kravis-roberts, poison-pill, kravis

CorporateFinance: acquisition, acquire, common-shares, common-stock, shares-outstanding, closely-held, preferred-stock, debentures, holding-company, common-shares-outstanding, initial-public-offering, definitive-agreement, preferred-shares, corporate-purposes, public-offering, buyback, quarterly-dividend, debenture

Earnings: cents, earnings, year-earlier, net-income, fourth-quarter, third-quarter, second-quarter, net-loss, revenue-rose, first-quarter, pretax, quarter-ended, operating-profit, continuing-operations, net-profit, second-quarter-earnings, year-earlier-quarter, analysts-expectations, fourth-quarter-earnings, per-share-earnings, diluted-share, profit-rose, first-quarter-earnings, analysts-estimates, profit-fell, fourth-quarter-net-income, pre-tax, year-ago-period, cents-per-share, third-quarter-results

ExecutiveCompensation: compensation, salary, bonus, bonuses, ceos, severance, stock-options

Financial: brokers, trades, traders, morgan-stanley, merrill, nyse, stock-exchange, brokerage, exchanges, nasdaq, merrill-lynch, ipo, brokerage-firm, trader, investment-banking, brokerage-firms, schwab, derivatives, arbitrage, volatility, merrill-lynch-co, stock-index, cboe, securities-dealers, trading-volume, underwriting, cme, charles-schwab, brokerages, commodity-futures-trading, initial-public-offerings

Foreign Exchange: dollar, currency, yen, euro, currencies, foreign-exchange, francs, japanese-yen, swiss-francs, british-pound, commodity-exchange, frankfurt, devaluation, comex, swiss-franc, franc, peso

Investing: portfolio, mutual-funds, hedge-funds, mutual-fund, portfolios, hedge-fund, pension-funds, asset-management, etf, etfs, morningstar, blackrock, emerging-markets, money-managers, pension-fund, fund-managers, exchange-traded, reits, bond-funds, portfolio-manager, capital-management, valuations, individual-investors, emerging-market, junk-bonds, fixed-income, reit, fidelity-investments, redemptions, bull-market, money-management, msci

Money Markets: deposit, certificates, treasury-bills, libor, prime-rate, commercial-paper, posted-yields, telerate, commercial-banks, discount-rate, treas, cds, commercial-paper-placed-directly, eu-rodollars, repurchase, interbank-offered, london-interbank-offered, treasury-notes, secondary-market, depository, lending-practices, c.d

Retail: stores, store, retailers, retail, retailer, sears, wal-mart, merchandise, chains, macy, department-stores, apparel, same-store, grocery, retailing, kmart, walmart, nordstrom, wal-mart-stores, home-depot, costco, catalog, supermarkets, merchandising, malls, groceries, kohls, lowe, kroger, roebuck, mart, drugstore, j.c-penney, in-store, zale, supermarket-chain, checkout, department-store, k-mart, penney

Securities: sec, accounting, exchange-commission, enron, audit, s.e.c, tyco, insider-trading, world-com, auditors, ebbers, auditor, financial-statements, audits, skilling, accountants, deloitte, kpmg, accounting-firm, bdo, auditing, fasb, restatement, securities-fraud, corporate-governance

Stock market: stocks, dow-jones-industrial-average, dow, stock-market, nasdaq-composite-index, standard-poor-500-stock, dow-industrials, blue-chip, nasdaq-composite, dow-jones-industrials, gainers, index-rose, vix, industrials, decliners, industrial-average, russell-index, composite-index, selloff, corporate-profits, investment-strategist, s-p-index, 10-year-treasury-note

Food Drinks

Beverage: brands, brand, coke, patent, coca-cola, pepsi, packaging, patents, beverage, pepsico, soft-drink, bottling, bottlers, soft-drinks, cola

Culinary: chef, menu, chicken, sauce, dishes, salad, cooking, cheese, meal, chocolate, soup, fried, vegetables, grilled, dish, pork, potatoes, shrimp, sandwich, steak, lamb, seafood, chefs, dessert, flavor, lobster, entrees, delicious, cuisine, culinary, appetizers, roast, sandwiches, desserts, spicy, roasted, toast, noodles, baked, sausage, salads, mushrooms, yogurt, brunch, spinach, peppers, chili, veal, creamy

Casual Dining: restaurant, restaurants, mcdonald, coffee, starbucks, dining, fast-food, grill, chipotle, diners, diner, cafes, taco, brew, deli, burgers, takeout, fries, applebee, dine, steakhouse, bagel, cream-cheese, dining-rooms

Wine: wine, beer, wines, bottle, drink, bottles, liquor, winery, grapes, tasting, grape, spirits, vineyard, vineyards, champagne, chardonnay, brewing, flavors, pinot-noir, liquors, beers, chateau, wine-spirits, burgundy, wineries, blanc, merlot, brewery, cork, drinkers, rum, bordeaux, sherry, chi-

anti, sonoma, sauvignon-blanc, pinot, cabernet, tasted, acidity, reds, tastings, winemaker, riesling, winemakers

Foreign Affairs

China: china, chinese, beijing, hong-kong, taiwan, shanghai, mainland, xinjiang, chinese-communist-party, taipei, wuhan, peking

Espionage: intelligence, cia, spy, classified, nsa, c.i.a, snowden, espionage, covert, spying, secrecy, intelligence-agencies, intelligence-committee, spies, wikileaks, classified-information, central-intelligence-agency, classified-documents, operatives, counterterrorism, clapper, leaking, briefings, intelligence-agency, encryption

Latin America: mexico, mexican, cuba, cuban, venezuela, argentina, castro, colombia, puerto-rico, chavez, latin-america, latin-american, nicaragua, colombian, venezuelan, argentine, havana, panama, dominican-republic, peru, el-salvador, puerto-rican, cubans, ecuador, buenos, guatemala, santiago

MiddleEastConflict: israel, israeli, palestinian, gaza, hamas, palestinians, arab, lebanon, egypt, middle-east, jerusalem, israelis, arafat, lebanese, beirut, egyptian, plo, cease-fire, arabs, palestine, aviv, hezbollah, cairo, israeli-military, gaza-strip, mideast, palestinian-authority, peace-talks, israeli-palestinian, syrians

Middle East: iraq, afghanistan, iraqi, syria, taliban, al-qaeda, afghan, islamic-state, pakistan, syrian, baghdad, libya, islamic, kabul, iraqis, pakistani, yemen, kurdish, sunni, saddam-hussein, bin-laden, insurgents, isis, libyan, kurds, insurgency, saddam, islamist, osama-bin-laden

Military: army, pentagon, navy, veterans, soldiers, commander, marines, naval, defense-secretary, marine-corps, national-guard, sgt, armed-forces, sergeant, colonel, generals, commanders, infantry, armed-services, active-duty, joint-chiefs, military-personnel, mattis, rumsfeld, admiral, battalion, troop, westmoreland, regiment, brig, naval-academy, cadets, hagel, saigon, deployments, lt-col, barracks, maj-gen, aircraft-carrier, servicemen, squadron

NuclearArms: nuclear, missiles, missile, nuclear-weapons, treaty, arms-control, cold-war, arsenal, chemical-weapons, warheads, disarmament, missile-defense, atomic, ballistic-missile, deterrence, ballistic-missiles, cruise-missiles, nonproliferation, trident, atomic-bomb

Russia & Ukraine: russia, russian, ukraine, moscow, ukrainian, kyiv, russians, kremlin, ukrainians, eastern-ukraine, russian-forces, crimea, separatists, russian-invasion, north-atlantic-treaty

Terrorism: terrorism, terrorist, terrorists, homeland-security, terrorist-attacks, cybersecurity, terrorist-attack, attackers, hijackers

Gender and Family

Abortion: abortion, abortions, pregnancy, abortion-rights, pregnant, reproductive, planned-parenthood, roe, antiabortion, embryos, roe-v-wade, pro-life, anti-abortion, bans, birth-control, fetus, pregnancies, contraception, pro-choice, unborn, pregnant-women, embryonic, fetal, incest

Gender: women, female, male, gender, sexual-harassment, feminist, harassment, sexually, masculinity, feminism, sexual-misconduct, sexism, masculine, sexist, feminine, feminists, macho, sexes

LGBTQ: gay, transgender, same-sex, lesbian, homosexual, lgbtq, gay-rights, trans, homosexuals, queer, sexuality, homosexuality, gays, sexual-orientation, lesbians, gender-identity, equality, heterosexual, bisexual, lgbt, gay-marriage

Marriage: marriage, wedding, couples, divorce, divorced, weddings, spouses, marry, romance, marriages, marital, honeymoon, husbands, married-couples, brides, divorces, unmarried

Parenting: children, parents, child, kids, mothers, babies, infant, fathers, parenting, parental, infants, foster-care, moms, children-ages, child-abuse, pediatrics, adolescents, toddlers, dads, 6-year-old, motherhood, adhd, pediatricians, adoptive

Gun

Gun_Control: gun, shooting, guns, firearms, gunman, shootings, rifle, gun-control, gun-violence, rifles, background-checks, handgun, massacre, pistol, firearm, nra, semiautomatic, sniper, handguns, gunfire, national-rifle-association, mass-shootings, mass-shooting, opened-fire, pistols, machine-guns, sandy-hook, gunshots

Health

Disease: surgery, brain, symptoms, transplant, kidney, diagnosed, diagnosis, tissue, liver, blood-pressure, syndrome, diseases, breast-cancer, organ, organs, heart-disease, heart-attack, disorders, arthritis, cancers, asthma, illnesses, chemotherapy, parkinson, transplants, tumor, cardiovascular, cardiac, inflammation, neurological, hypertension, coronary, prostate-cancer, kidneys, clots, angioplasty, scans, tumors, blood-vessels, cardiologist, cancer-patients

Healthcare: insurance, coverage, medicare, insurers, health-insurance, medicaid, premiums, insurer, insurance-companies, aetna, uninsured, insured, reinsurance, blue-cross, cigna, policyholders, unitedhealth, out-of-pocket, blue-shield, hmos, deductibles, long-term-care, insurance-coverage, health-care-costs, managed-care, single-payer, hmo, insurance-premiums, humana, reimbursements

Hospital: hospital, medical, patients, doctors, hospitals, patient, doctor, physicians, physician, nurses, clinic, nurse, nursing-home, healthcare, nursing-homes, medical-care, dental, nursing, emergency-room, surgeons, surgical, outpatient, practitioners, hospice, intensive-care, primary-care, orthopedic, telehealth, paramedics, geriatric, internist, anesthesia, oncology, health-care-providers

MentalHealth: depression, anxiety, mental-health, trauma, psychiatrist, therapist, psychologist, psychologists, empathy, psychiatrists

Nutrition: foods, diet, fat, nutrition, obesity, calories, diabetes, insulin, supplements, protein, vitamin, weight-loss, overweight, dietary, vitamin-d, cholesterol, blood-sugar, diets, nutritional, obese, cereal, intake, sodium, fruits, vitamins, glucose, grams, calorie, fats, bran, caffeine, grains, oat, cereals, low-fat, sugary, nutrients, type-diabetes, calcium, peanut, allergy, fatty, additives, sugars, healthful, grapefruit, nutrient

Pharmaceuticals: drugs, fda, pfizer, drug-administration, merck, pharmaceutical, pharmaceuticals, f.d.a, trials, pharmacy, medicines, treatments, prescription, medications, prescriptions, clinical-trials, side-effects, pill, biotech, azt, pharmaceutical-companies, prescription-drugs, pharmacies, biogen, genentech, alzheimer, astrazeneca, over-the-counter, eli-lilly, clinical-trial

Substance Abuse: drug, marijuana, cocaine, alcohol, addiction, heroin, steroids, cannabis, opioid, addicts, overdose, narcotics, drug-abuse, fentanyl, cbd, steroid, addict, alcoholism, substances, opioids, substance-abuse, painkillers, alcoholics, rehab, trafficking, thc, methadone, sobriety, overdoses, drug-related

Vaccine: vaccine, virus, aids, vaccines, flu, infected, infection, vaccination, cdc, hiv, infections, outbreak, disease-control, vaccinated, immune, viruses, epidemic, outbreaks, antibodies, measles, vaccinations, infectious-disease, influenza, variant, fauci, infectious-diseases, c.d.c, h.i.v, variants, contagious, immune-system, unvaccinated, aids-virus, antibody, public-health, smallpox

Home

Cooking: salt, pepper, tablespoons, butter, teaspoon, recipe, garlic, cups, oven, tomatoes, mixture, flour, olive-oil, chopped, tablespoon, baking, beans, onion, skillet, lemon, pasta, cooked, recipes, parsley, bake, boil, minced, freshly-ground, simmer, onions, tomato, spoon, black-pepper, grated, teaspoons, sprinkle, basil, tofu, peeled, eggplant, thyme, garnish, lemon-juice, saucepan

Home: kitchen, furniture, bedrooms, living-room, bathrooms, bathroom, dining-room, fireplace, renovated, ceilings, furnishings, cabinets, fireplaces, furnished, tub, amenities, flooring, tiles, decorating, ikea, interior-designer, wallpaper, foyer, closets, hardwood, staircase

Household: plastic, carpet, cardboard, towels, fibers, stain, dryer, cleaners, gel, stains, polyester, acrylic

Housing

Architecture: tower, architecture, floors, towers, architects, architectural, elevator, exterior, stairwell

Housing: rent, apartments, rental, homeowners, tenants, condominium, co-op, rents, hud, builders, landlords, tenant, landlord, condo, builder, affordable-housing, realtors, renters, renting, home-

builders, two-bedroom, condominiums, co-ops, single-family, urban-development, rentals, home-ownership, condos, one-bedroom, single-family-homes, eviction, apartment-buildings, complexes, dwellings, townhouses

Mortgage: loans, mortgage, loan, mortgages, lenders, borrowers, lending, lender, fannie-mae, freddie-mac, subprime, freddie, fannie, foreclosure, borrowing, refinance, borrower, foreclosures, refinancing, fha, appraisal, monthly-payments, mortgage-rates, countrywide, defaults, repayment, home-equity, mortgage-backed-securities, 30-year-fixed, appraisals, appraisers, fixed-rate, refinanced

Urban development: developers, developer, square-feet, zoning, renovation, square-foot, redevelopment, mixed-use, high-rise, rezoning, land-use, lofts, multifamily, shopping-centers, walkable

Infrastructure

Electricity: plant, utility, utilities, electricity, edison, reactor, grid, nuclear-power, transmission, reactors, uranium, nuclear-plant, pepco, electric-power, generators, power-plant, nuclear-power-plant, seabrook, blackout, outage, nuclear-power-plants, outages, generator, megawatts, turbines, hydroelectric

Energy: oil, gas, energy, gasoline, natural-gas, petroleum, pipeline, drilling, exxon, oil-prices, barrel, opec, shell, crude, exploration, chevron, texaco, ethanol, barrels, bp, refinery, mobil, million-barrels, refining, crude-oil, shale, refineries, pennzoil, refiners, fuels, pipelines, fracking, gasoline-prices, arco, exxon-mobil, gas-prices, amoco, rigs, unocal, royal-dutch-shell, petroleum-exporting-countries

Postal service: mail, postal-service, postal, stamp, stamps, mailing, mailed, envelope, p.o, postage, postmaster, mailbox

Telecommunications: cable, wireless, telecommunications, verizon, telecom, fcc, sprint, cellular, long-distance, mci, comcast, motorola, broadband, time-warner, qwest, nokia, communications-commission, cable-television, bell-atlantic, cablevision, wi-fi, ericsson, bellsouth, cable-tv, t-mobile, fiber-optic, lucent, nextel, sbc, cisco, 5g, nynex, 3g, cable-operators, ameritech, verizon-wireless, mobile-phone, nortel

Water: snow, ice, drought, freezing, dam, pipes, sewer, sewage, reservoir, frigid, basin, dams, drinking-water, reservoirs, melting, moisture, snowstorm, downstream

Legal Institution

Constitutional Law: amendment, statute, first-amendment, statutes, constitutionality, equal-protection

Judicial: judge, ruling, appeals, judges, courts, appeals-court, federal-judge, upheld, circuit-court, appellate, district-judge, federal-appeals-court, rulings, unconstitutional, court-ruling, injunction, federal-courts, court-ruled, three-judge, supreme-court-ruled, judge-ruled, oral-arguments

Lawsuit: suit, settlement, lawsuit, lawsuits, litigation, sued, plaintiffs, damages, antitrust, ftc, attorneys, district-court, settlements, arbitration, alleges, suing, class-action, lawsuit-filed, suit-filed, plaintiff, punitive-damages, legal-fees, attorneys-general, trade-commission, filed-suit, defamation, trial-lawyers, deposition, libel, restitution, eeoc, tort, suit-alleges, class-action-lawsuit, f.t.c, consent-decree

SupremeCourt: justice, supreme-court, roberts, justices, judicial, constitution, confirmation, constitutional, scalia, ginsburg, clarence-thomas, kavanaugh, judge-kavanaugh, judiciary, confirmation-hearings, precedents, jurisprudence, senate-judiciary-committee, constitutional-right, federalism

Leisure

Festival: festival, parade, fireworks, gala, festivities, ceremonies, festivals, fest

Gaming: gambling, casino, las-vegas, casinos, gaming, vegas, gamblers, gambler, roulette

Horticulture: garden, tree, trees, flowers, flower, planted, leaf, seeds, varieties, roses, planting, gardeners, bulbs, bees, foliage, gardening, gardener, bulb, insect, blooms, flowering, blossoms, pollen, shrubs, watering, grasses, seedlings, botanic, blooming, compost, botanical

Nature: lake, mountain, trail, ski, mountains, trails, hike, hiking, skiing, canyon, skis, skiers, slopes, fir, cliffs, camping, vail, skier

TravelIndustry: hotel, travel, hotels, travelers, resort, rooms, hilton, marriott, tourism, spa, hospitality, traveler, booked, resorts, tourist, motel, lodging, booking, occupancy, hyatt, bookings, hotel-rooms, alamo, inns, ritz-carlton, itinerary, spas, check-in, concierge, wyndham

Media

Journalism: magazine, newspaper, editor, reporter, newspapers, publisher, journalists, journalism, readers, editors, magazines, articles, publications, publishing, editorial, circulation, tribune, managing-editor, front-page, newsweek, herald, newsroom, news-organizations, daily-news, tabloid, journalistic, usa-today, gannett, correspondents, sports-illustrated, newsday, chicago-tribune, pulitzer, editorial-page, pulitzer-prize, boston-globe, inquirer

Radio: radio, tape, stations, broadcasting, tapes, radio-stations, broadcasters, radio-station, broadcasts, broadcaster, airwaves, cassette

TV Entertainment: episode, comedy, episodes, hbo, comedian, premiere, pbs, stand-up, podcast, executive-producer, late-night, sitcom, showtime, reality-tv, comedians, bravo, emmy, premieres, tv-shows, satire, co-star

Television: network, cbs, nbc, abc, viewers, broadcast, cnn, anchor, nielsen, prime-time, msnbc, mtv, cbs-news, nbc-news, wb, telecast, viewer, couric, anchors, brokaw, viewership, on-air, koppel, miniseries

Outer Space

Space: earth, nasa, moon, shuttle, mars, rocket, planet, astronauts, spacecraft, orbit, satellites, space-station, apollo, space-shuttle, astronaut, capsule, jupiter, rockets, rover, planets, telescope, lunar, astronomers, comet, aeronautics, observatory, orbital, astronomy, orbiting, planetary, propulsion, capsules, voyager, comets

Politics

Campaigns Donors: donors, donations, contributions, donated, fund-raising, campaigns, lobbying, donor, donation, charities, campaign-finance, donate, lobbyists, foundations, fundraising, pac, fund-raiser, fec, pacs, super-pac, donating, campaign-contributions, fund-raisers

Elections: election, voters, voting, votes, ballot, elections, ballots, voter, voting-rights, turnout, recount, referendum, voter-registration, electoral, electoral-votes, mail-in, absentee, electors, precincts, balloting, election-results, voter-fraud, suppression, electoral-college, voting-rights-act, canvassing, term-limits, exit-polls

Protest: protest, protesters, protests, activists, demonstrations, demonstrators, peaceful, riot, activism, protesting, free-speech, tear-gas, police-brutality, racial-justice, marchers, protester

PublicOpinion: survey, poll, respondents, surveys, surveyed, polling, polled, pew, gallup, pew-research, registered-voters, pollsters, approval-rating, quinnipiac

Senate: bill, senate, legislation, senator, senators, bipartisan, veto, majority-leader, caucus, senate-majority-leader, filibuster, amendments, committee-chairman, senate-floor, minority-leader, earmarks, congressional-leaders, appropriations-committee, republican-senators, senate-finance-committee, democratic-caucus

Race and Immigration

Confederate: monument, confederate, slaves, monuments, gettysburg, confederacy

Immigration: immigration, border, immigrants, migrants, citizenship, immigrant, refugees, asylum, visas, migration, border-patrol, illegal-immigrants, migrant, refugee, deportation, haitian, asylum-seekers, undocumented, southern-border, undocumented-immigrants, deported, border-protection, illegal-immigration, cbp, haitians, passports, naturalization, immigration-reform, customs-enforcement, deport, smugglers, illegal-aliens, dhs, unaccompanied

Minority: diversity, hispanic, discrimination, minorities, ethnic, affirmative-action, latino, hispanics, latinos, asians, disparities, ethnicity, minority-groups, civil-rights-act, discriminated, underrepresented

Native American: indian, indians, tribes, tribal, tribe, reservation, native-american, native-americans, indigenous, sioux, navajo

Racial: black, blacks, african-american, racial, civil-rights, whites, racism, african-americans, racist, slavery, slave, reparations, negro, martin-luther-king-jr, racially, naacp, segregated, segregation, civil-rights-movement, klan, desegregation, farrakhan, klux, enslaved, busing, jesse-jackson, sharpston, predominantly-black, black-owned, confederate-flag, lynching, malcolm-x, africans, supremacist

Religion

Catholicism: church, pope, churches, catholic, rev, pastor, bishop, vatican, cardinal, holy, priest, archdiocese, bishops, priests, baptist-church, congregation, gospel, parish, benedict, archbishop, catholics, worship, diocese, catholic-church, clergy, paul-ii, nuns, protestant, mormon, roman-catholic, parishioners, pope-john, pastors, pope-francis, parishes, sermon, jesus-christ, theology, papal, sermons, congregations, theological, communion, missionary, roman-catholic-church, messiah

Religion: religious, jewish, christian, god, religion, jews, muslim, bible, rabbi, prayer, muslims, jesus, christians, islam, evangelical, synagogue, orthodox, secular, prayers, christianity, biblical, evangelicals, judaism, mosque, hebrew, jew, jewish-community, religious-freedom, yiddish, rabbis, buddhist, religions, religious-beliefs, anti-semitism, passover, monks, faith-based, ultra-orthodox, spirituality, mosques, protestants

Science

Science: scientists, scientific, cells, gene, dna, genetic, genes, laboratory, biological, bacteria, biology, physics, mice, genetics, proteins, genome, molecules, genetically, evolutionary, molecular, einstein, rna, biologists, human-genome, organisms, hypothesis, nobel-prize, physicists, biologist, microbes, mutations, bacterium

Sports

Baseball: baseball, major-league-baseball, mlb, ballpark, umpires, babe-ruth, yankees, mets, innings, orioles, pitcher, inning, pitching, pitched, home-runs, starter, red-sox, major-league, pitchers, pitches, catcher, homer, bullpen, fastball, phillies, braves, reliever, homers, rbi, spring-training, hitter, batters, mound, blue-jays, white-sox, right-hander, batting, two-run, homered

Boxing: boxing, ali, tyson, wrestling, boxer, bout, heavyweight, punches, jab, muhammad-ali, welterweight, bouts, boxers, wrestler, knockout

CollegeSports: coaches, ncaa, freshman, coaching, notre-dame, acc, n.c.a.a, virginia-tech, big-ten, ohio-state, college-football, ncaa-tournament, head-coach, athletic-director, clemson, coached, penn-state, college-basketball, rutgers, athletics, football-coach, championship-game, assistant-coach, u.c.l.a, wake-forest, basketball-coach, villanova, hokies, men-basketball, tar-heels, georgia-tech, atlantic-coast-conference, collegiate-athletic-association, krzyzewski, jayhawks, coach-mike,

longhorns, student-athletes

Golf: golf, hole, holes, tee, golfers, pga-tour, greens, putt, golf-course, golf-club, birdie, strokes, tiger-woods, birdies, bogey, golfer, pga, final-round, fairway, ryder-cup, putts, birdied

NBA: nba, knicks, nets, lakers, warriors, n.b.a, bulls, celtics, pistons, clippers, 76ers, golden-state, spurs, suns, point-guard, mavericks, bucks, nuggets, pacers, wnba, eastern-conference, basketball-association, los-angeles-lakers, arenas, boston-celtics, mvp

NFL: redskins, nfl, quarterback, jets, super-bowl, linebacker, cowboys, raiders, rams, chiefs, eagles, wide-receiver, n.f.l, ravens, cornerback, browns, 49ers, packers, nfc, quarterbacks, preseason, afc, chargers, steelers, colts, tight-end, defensive-end, dolphins, green-bay, sacks, offensive-line, titans, training-camp, broncos, bengals, linemen, tackles

NHL: rangers, islanders, hockey, puck, devils, nhl, goalie, defenseman, third-period, flyers, penguins, goaltender, n.h.l, stanley-cup, bruins, power-play, canadiens, red-wings, blackhawks, washington-capitals, shutout, oilers, sabres, maple-leafs, messier, gretzky, national-hockey-league, canucks, shootout, goalies

Soccer: soccer, world-cup, mls, midfielder, fifa, goalkeeper, soccer-team, striker, midfield, fc

Tennis: tennis, tournament, singles, wimbledon, doubles, semifinals, grand-slam, quarterfinals, tournaments, quarterfinal, racket, defending-champion, backhand, tiebreaker, forehand

Transportation

Airline: airlines, airline, airport, flight, plane, flights, planes, aircraft, boeing, pilots, delta, carrier, carriers, aviation, airports, pilot, jet, continental, faa, routes, american-airlines, airplane, federal-aviation-administration, international-airport, usair, united-airlines, us-airways, airways, flown, airplanes, f.a.a, delta-air-lines, twa, air-traffic, flight-attendants, british-airways, airline-industry, concorde, northwest-airlines, o'hare

Automotive: ford, gm, chrysler, vehicles, auto, toyota, general-motors, auto-makers, auto-maker, honda, g.m, volkswagen, automotive, nissan, auto-industry, ford-motor-co, general-motors-corp, vw, motors, automakers, chevrolet, sport-utility, car-makers, mitsubishi, iacocca, audi, electric-vehicles, hyundai, volvo, automaker, dealership, diesel, fiat, ford-motor, saturn, pickups, mazda, mulally, dealerships, toyota-motor, pontiac, minivans, evs, jeep

Maritime: ship, boat, ships, boats, cruise, aboard, vessels, vessel, coast-guard, cargo, maritime, sailing, yacht, ports, submarine, voyage, dock, sailor, sailors, seas, tanker, battleship, sailed, hull, crew-members, cruises, wharf, submarines, cruise-ship, shipbuilding, uss, yachts, shipyard, sails, skipper, strait, destroyer, barge, shippers

Mass Transit: metro, bus, transit, subway, trains, buses, commuters, ridership, metrorail, commuter, mta, transit-authority, m.t.a, transit-system, metropolitan-transportation-authority, metrobus, commute, mass-transit, light-rail, subways, streetcar

Railroad: train, railroad, rail, amtrak, penn, railroads, freight, csx, railway, union-pacific, norfolk-southern, conrail, burlington-northern, derailment, locomotive, railways

Traffic: traffic, bridge, highway, lanes, parking, toll, bridges, highways, congestion, beltway, motorists, pedestrians, tolls, roadway, rush-hour, expressway, potholes, intersections, traffic-congestion, roadways, ramps, interchange, transportation-authority, parking-spaces

Transportation: car, cars, driver, driving, drivers, engine, taxi, wheel, wheels, bmw, mileage, brake, brakes, mercedes, horsepower, towing, taxis, suv, steering-wheel, cadillac, dashboard, sedan, minivan, ferrari, porsche, air-bags, acceleration, seat-belt, subaru, cabs, drunken-driving, hummer, chevy, highway-traffic, seat-belts, m.p.h, chassis, lexus, miles-per-gallon

Appendix E: Constructing Confidence Intervals

I followed the subsampling approach used by Kozlowski et al. (2019: 934-935) and partitioned my corpus into 20 random subsamples. For each subsample k , I applied à la carte word embedding models and calculated the cosine similarity s_i^k between the embedding vector of a given term i and the embedding vectors of ideological/partisan labels in each subsample, which captured the extent to which the two vectors are contextually similar. I used the average of the 20 cosine similarity measures, $\bar{s}_i$, as the measurement for cosine similarity for term i . I then calculated the error of the cosine similarity estimate in each subsample k , E_i^k , as $\sqrt{\tau_k}(s_i^k - \bar{s}_i)$, where τ_k is the number of texts in the k_{th} sample and s_i^k is the cosine similarity for the k_{th} sample. I used the 5th percentile and 95th percentile values of the errors, adjusted for the number of total texts, to construct the upper and lower bounds of the 90 percent confidence interval for the cosine similarity estimate. Formally, the upper bound is $\bar{s}_i + \frac{E_{i,0.95}}{\sqrt{\tau}}$, and the lower bound is $\bar{s}_i - \frac{E_{i,0.05}}{\sqrt{\tau}}$, where $\bar{s}_i$ is the mean of the 20 cosine similarity estimates for a given term i and τ is the number of texts in the whole corpus, $E_{i,0.05}$ is the 5th percentile of the 20 error estimates for term i , and $E_{i,0.95}$ is the 95th percentile error. I then determined the 90% confidence interval for the issue-level politicization scores by averaging the upper and lower bounds of each dictionary term's confidence interval in the dictionary.

Appendix F: News-Only Robustness Check

The main analysis in the paper includes both news articles and editorials. To check whether the results change when editorials and commentaries are excluded, I re-ran the analysis with articles whose article types are “News,” “Article,” “Feature,” “Front Page,” or “Cover Story” in the metadata and excluded editorials, commentaries, and reviews.

Figure F1 shows that the trajectory of politicization is similar across the two samples, with a comparable trough in the early 1990s. The two series diverge most in the pace of politicization that follows: through the late 1990s and early 2000s, the all-articles trend rises much faster in politicization than the news-only trend, suggesting that the initial resurgence of politicized language appeared most strongly in opinion content. From the mid-2000s onward, the news-only series accelerates and narrows much of that gap, though a modest difference between the two persists through 2024. This suggests that opinion content might be leading the increase in politicized discourse in the 1990s, with news coverage substantially catching up over the following two decades.

Figures F2 and F3 replicate the original Figures 4 and 5, showing domain-specific politicization for the ideology and party respectively. The pattern of increase from the 1980–1989 average to the 2015–2024 average is very similar across the All Articles and News-Only samples: the same set of domains — Race and Immigration, Legal Institution, Gender and Family, Religion, and Education — is the most politicized in the 2015–24 period in both the original and news-only analyses. The relative order across domains is also largely preserved: rank positions shift by no more than 3 places between the two samples. Domains that are the least politicized (Home, Outer Space, Financial Market) remain so in both samples. The main difference is that the increase in politicization is somewhat smaller on average in the news-only sample.

Taken together, these results indicate that the generalized upward drift in politicization documented in Figure 4 is not an artifact of the inclusion of opinion content. The primary consequence of including editorials and commentaries appears to be an earlier and somewhat steeper politicization trajectory.

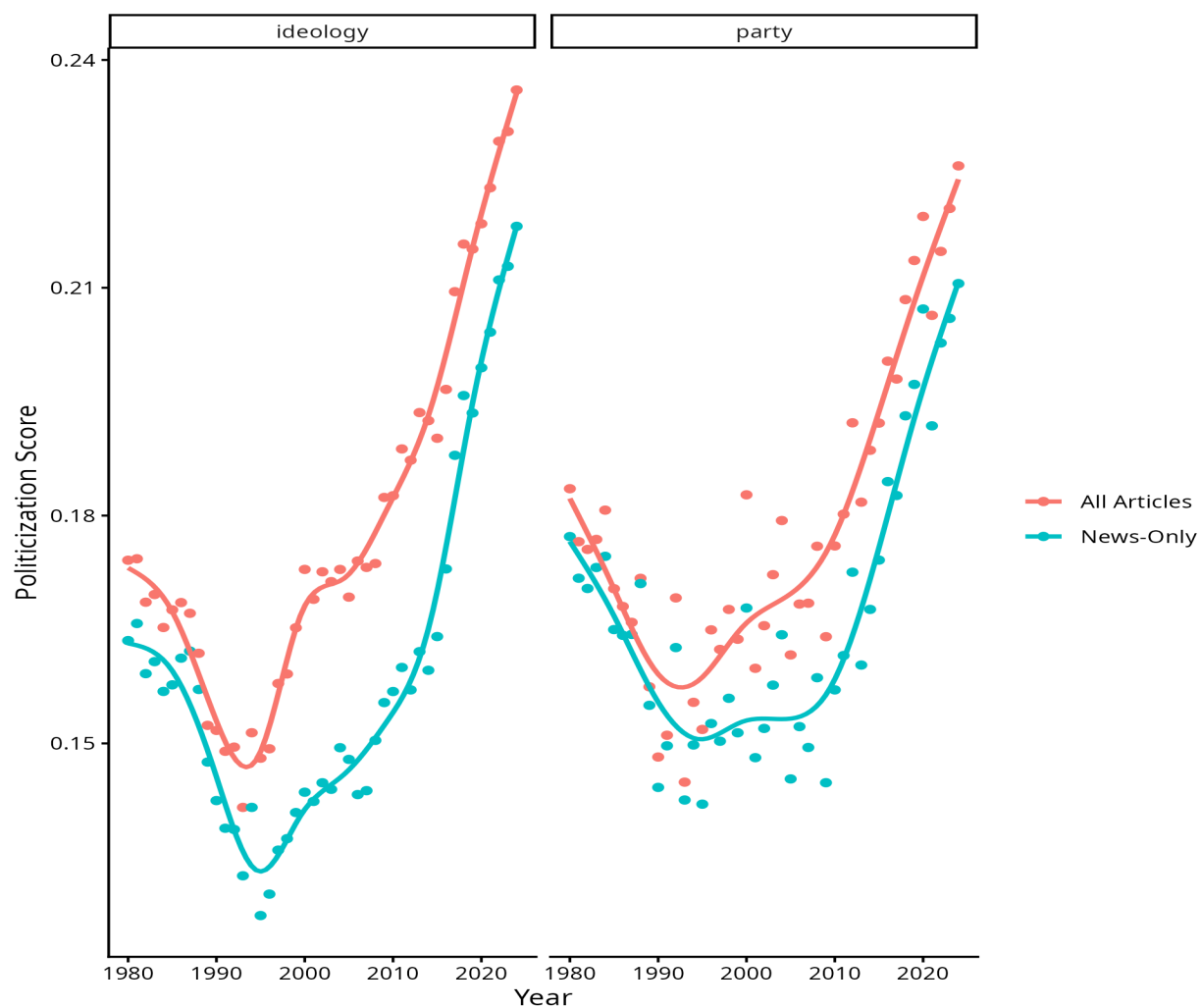


Figure F1. Average Ideological and Partisan Politicization, 1980–2024; All Articles vs. News Only
Note: Trend lines are smoothed with a generalized additive model.

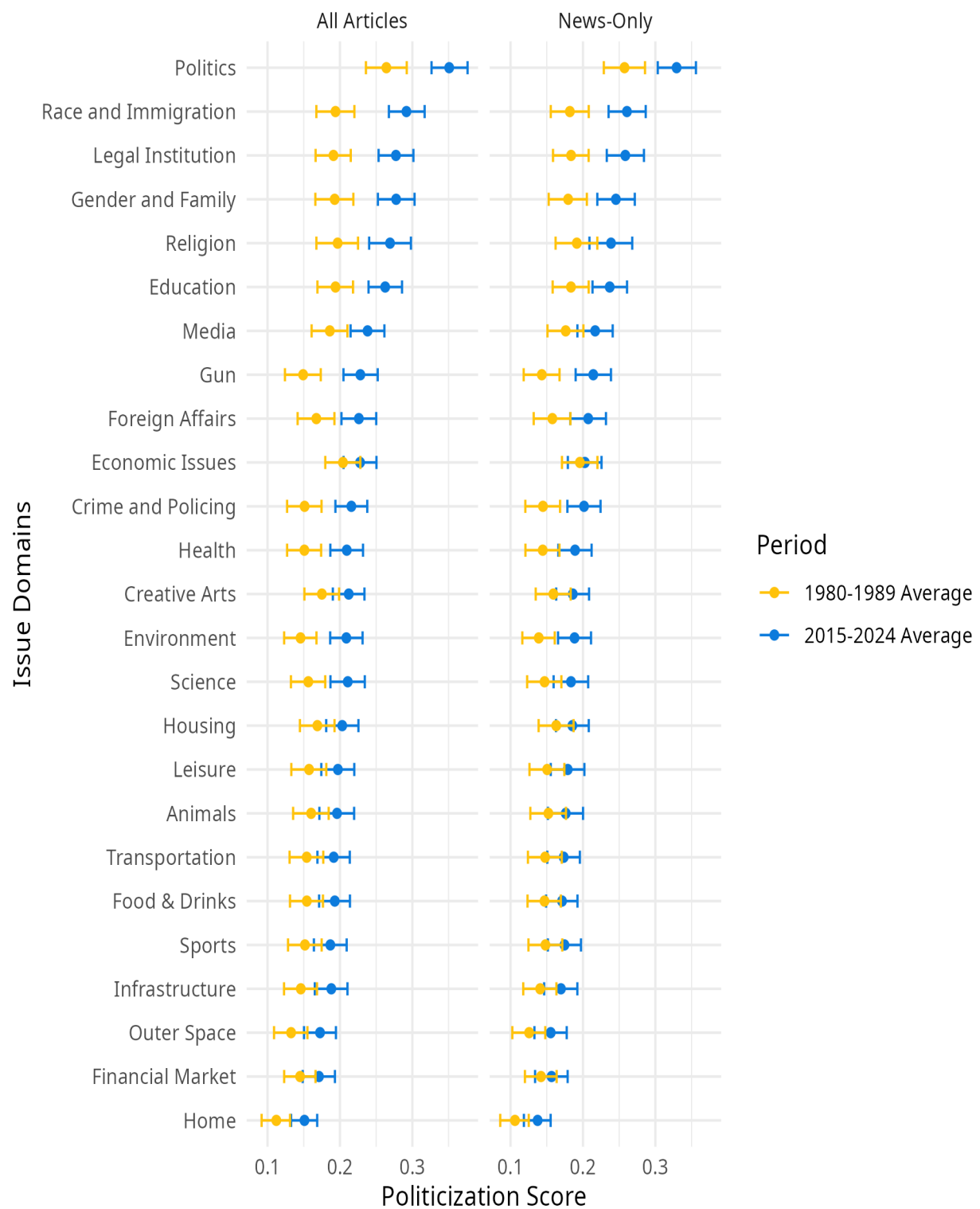


Figure F2. Changes in Ideological Politicization Score by Issue Domains, Period Comparison; All Articles vs. News Only

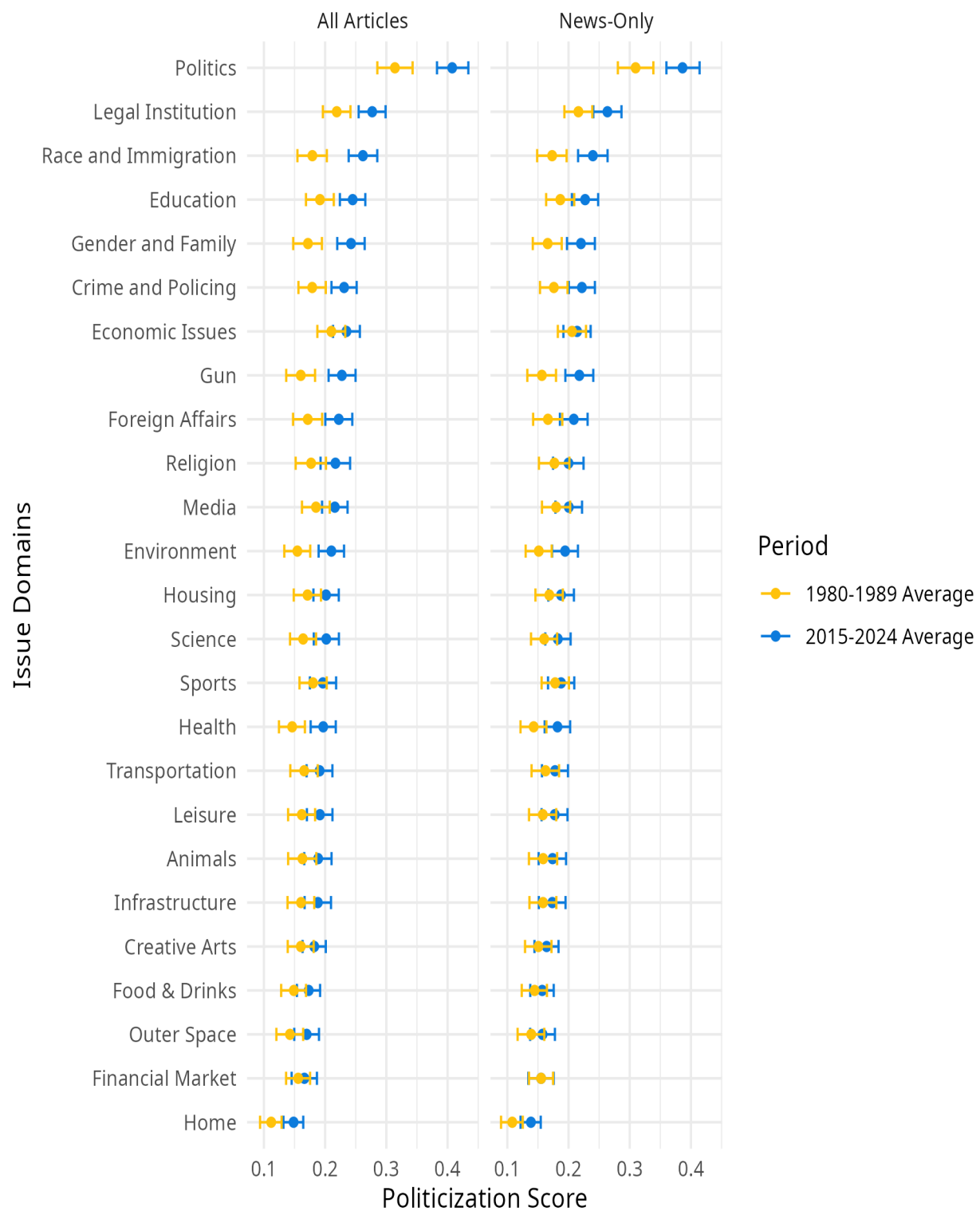


Figure F3. Changes in Partisan Politicization Score by Issue Domains, Period Comparison; All Articles vs. News Only

Appendix G: Which Newspapers Drive Politicization?

To investigate whether patterns of ideological politicization vary between the New York Times, Wall Street Journal, and Washington Post, I conducted a separate analysis for each newspaper. I ran separate à la carte embedding models for each outlet and constructed politicization scores using outlet-specific average embedding vectors.

Table G1 replicates the multilevel model presented in Table 1, now predicting levels of politicization by year separately for the three newspapers.¹ The findings indicate that the New York Times contributed the most to increasing politicization over the last forty years. In particular, the New York Times exhibited an average annual increase of 0.0017 in ideological politicization across the topics analyzed ($p < 0.001$). The Washington Post and The Wall Street Journal also saw a statistically significant increase in ideological politicization ($p < 0.001$) since 1980, albeit with a smaller effect size than The New York Times (0.0008 and 0.0012, respectively). Table G2 shows similar findings for partisan politicization.

Table G1: Multilevel Model Predicting Levels of Ideological Politicization by Year, Separated by Outlet

	New York Times	Washington Post	Wall Street Journal
Fixed effect:			
Intercept	0.0878*** (0.0019)	0.1159*** (0.0023)	0.1013*** (0.0020)
Year	0.0017*** (0.0001)	0.0008*** (0.0001)	0.0012*** (0.0000)
AIC	-27137.8009	-28341.5039	-31281.0774
BIC	-27097.7128	-28301.4158	-31240.9893
Log Likelihood	13574.9004	14176.7520	15646.5387
Num. obs.	5892	5892	5892
Num. groups: topic	131	131	131
Var: topic Intercept	0.0004	0.0007	0.0005
Var: topic Year	0.0000	0.0000	0.0000
Cov: topic Intercept Year	0.0000	0.0000	0.0000
Var: Residual	0.0005	0.0004	0.0002

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Note: This is a random-intercept and random-slope model. The year variable is zeroed at 1980. Numbers in parentheses are standard errors.

¹Terms must appear at least 5 times for each newspaper and at least 10 times across the three outlets in a given year to be included in the analysis.

Table G2: Multilevel Model Predicting Levels of Partisan Politicization by Year, Separated by Outlet

	New York Times	Washington Post	Wall Street Journal
Fixed effect:			
Intercept	0.1128*** (0.0030)	0.1409*** (0.0031)	0.1116*** (0.0030)
Year	0.0013*** (0.0001)	0.0005*** (0.0001)	0.0011*** (0.0001)
AIC	−28002.7433	−27804.6161	−30333.3154
BIC	−27962.6552	−27764.5280	−30293.2273
Log Likelihood	14007.3717	13908.3081	15172.6577
Num. obs.	5892	5892	5892
Num. groups: topic	131	131	131
Var: topic Intercept	0.0012	0.0012	0.0011
Var: topic Year	0.0000	0.0000	0.0000
Cov: topic Intercept Year	−0.0000	0.0000	0.0000
Var: Residual	0.0004	0.0004	0.0003

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Note: This is a random-intercept and random-slope model. The year variable is zeroed at 1980. Numbers in parentheses are standard errors.

Turning to specific issues, Figure G1 and G2 present a comparison of the average ideological and partisan politicization scores for the 25 issue domains between the 1980-1989 period and the 2015-2024 period for each newspaper. In the figures, arrows represent the change in politicization for each issue over time, with the starting point of each arrow marking the average politicization score in the first period and the endpoint marking the score in the second period. The results suggest that, for race and immigration, legal institutions, gender and family, education, and many other issues, all three outlets show a clear increase in politicization between the two periods. However, there are also some issues where only the New York Times showed much more increase in politicization than the other two outlets, such as media and economic issues.

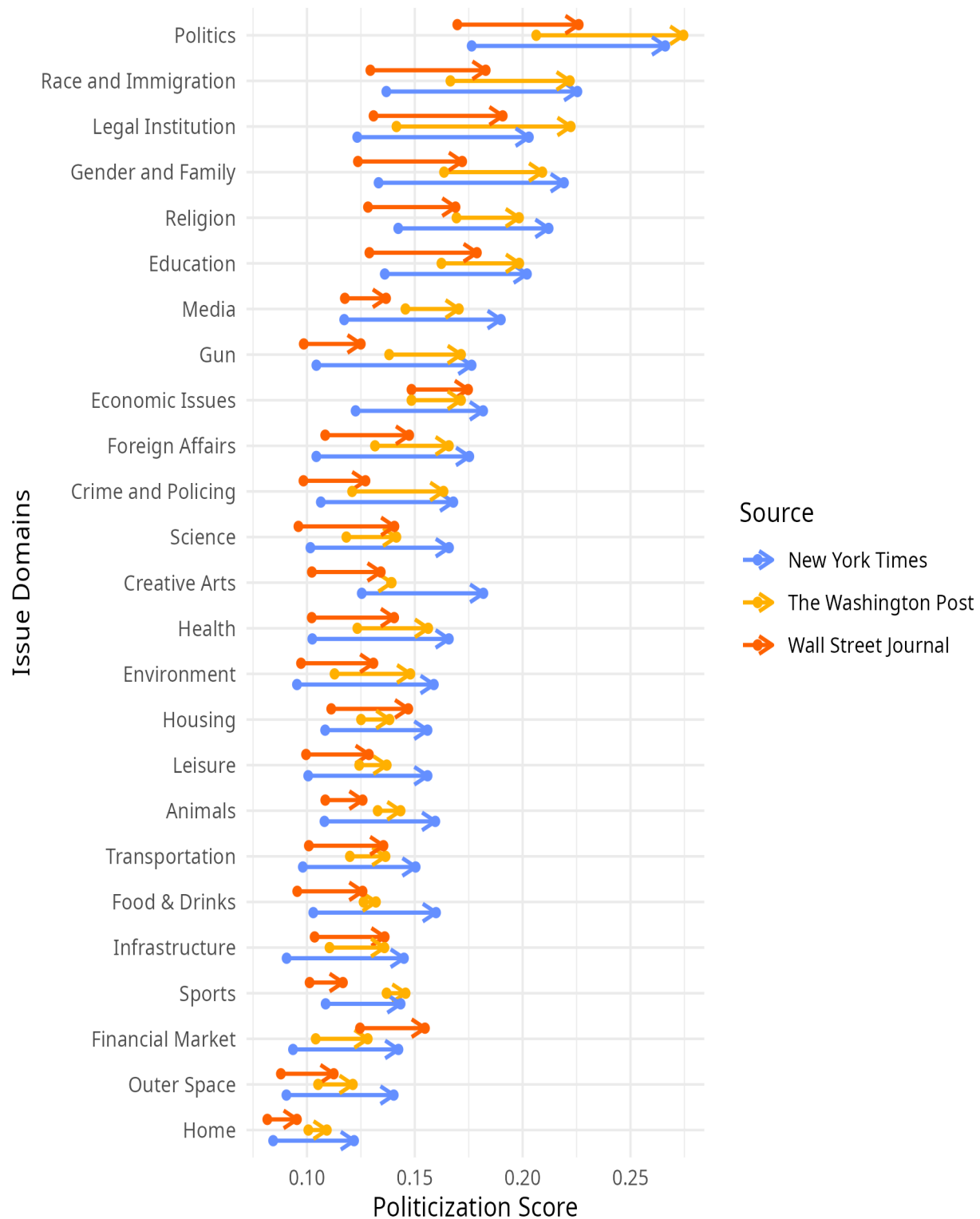


Figure G1. Changes in Ideological Politicization Score by Issue Domains, Period Comparison; Separate Analysis on 3 National Newspapers

Note: The starting point of each arrow marks the average politicization score in the 1980-1989 period, and the endpoint marks the score in the 2015-2024 period.

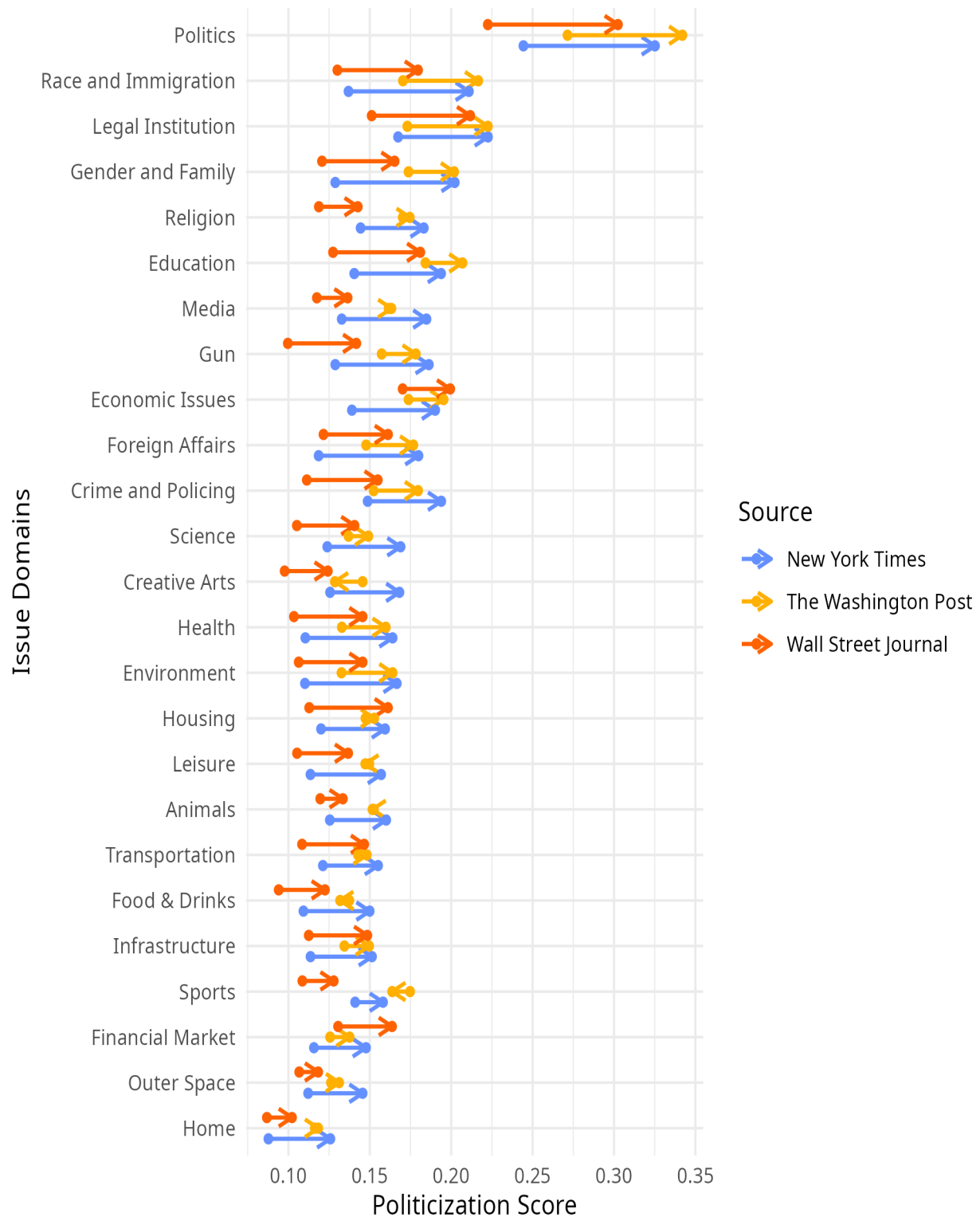


Figure G2. Changes in Partisan Politicization Score by Issue Domains, Period Comparison; Separate Analysis on 3 National Newspapers

Note: The starting point of each arrow marks the average politicization score in the 1980-1989 period, and the endpoint marks the score in the 2015-2024 period.

Appendix H: Politicization of Most Unique Terms in Lifestyle Topics

Figure 8 in the main text illustrates term-level politicization trends using 20 terms selected by the author. To address the possibility that this selection could overstate or understate politicization in these domains, this appendix instead selects terms using an objective rule: for each of the 25 topics within Creative Arts, Media, Sports, and Food & Drinks, I select the single term with the highest uniqueness score (the same score used to construct the topic dictionaries; see Equation 1).

Figure H1 plots the resulting 25 terms. Of these, 11 show a significant increase in ideological politicization from 1980–1989 to 2015–2024, including *disney*, *world-cup*, *comedy*, *musicals*, and *innings*. The remaining terms show no significant change, and one term, *redskins*, shows a significant decrease (although this decline coincides with the team’s 2020 name change and is not necessarily evidence of depoliticization). Overall, this term-level pattern is consistent with the pattern reported in Figure 8: a minority of lifestyle terms show a clear politicization increase, while most do not.



Figure H1. Term-Level Ideological Politicization in Popular Culture, 1980–1989 Average and 2015–2024 Average; Most Unique Term per Topic

Note: For each of the 25 topics within Creative Arts, Media, Sports, and Food & Drinks, the term with the highest uniqueness score (Equation 1) is shown. Error bars show the 90% confidence interval, calculated by averaging the upper and lower bounds of the 90% confidence interval of the term-year politicization scores within each 10-year period. Terms are sorted by the difference in politicization between 2015–2024 and 1980–1989.

Appendix I: ANES Items Used and Corresponding Issues

I matched issues in my politicization dataset with items in the ANES Time Series Cumulative Data File (<https://electionstudies.org/data-center/anes-time-series-cumulative-data-file/>). I reviewed all ANES questions and retained those that most closely correspond to the underlying issue dimensions represented by topics in my politicization dataset. In addition to the items used by Kozlowski and Murphy (2021), I also included items such as “Increase or decrease number of immigrants to U.S.” (matched to Immigration) and “How Much Does the Federal Government Waste Tax Money” (matched to Taxation). Not all ANES items are included because some are too broad to map onto specific topics in the politicization dataset (e.g., “Tolerance of different values” is excluded). Below, I list the 15 issues identified through this procedure, the corresponding topics in the politicization dataset, and the associated ANES questions.

1. Abortion

Politicization: Abortion

ANES: When should abortion be allowed vcf0837

2. Crime

Politicization: CriminalJustice, Police

ANES: Federal spending on crime vcf0888

3. Education

Politicization: Education

ANES: Federal spending on public schools vcf0890

4. Environment

Politicization: Air Pollution, ClimateChange, Disaster, Wildfires

ANES: Environment Regulation Scale vcf0842; Federal spending on the environment vcf9047

5. Gender

Politicization: Gender

ANES: Women equal role scale vcf0834

6. Healthcare

Politicization: Healthcare

ANES: Government health insurance vcf0806

7. Immigration

Politicization: Immigration

ANES: Increase or decrease number of immigrants to U.S. vcf0879

8. LGBTQ

Politicization: LGBTQ

ANES: Homosexuals in military vcf0877; Laws to protect homosexuals vcf0876a; Homosexuals allowed to adopt children vcf0878

9. Military

Politicization: Military

ANES: Defense spending vcf0843

10. Race

Politicization: Racial, Minority

ANES: Affirmative action vcf0867a; Blacks got less than they deserved vcf9042; Government should ensure fair jobs for Blacks vcf9037; Amount of government aid to Blacks vcf0830; Blacks should not have favors vcf9040; School integration vcf0816; Conditions make it hard for blacks to succeed vcf9039; Blacks must try harder vcf9041; Does civil rights push too fast vcf0814; How much has position of blacks changed vcf0813

11. Religion

Politicization: Catholicism, Religion

ANES: Authority of the Bible vcf0850; Prayer in schools vcf9043

12. Space and Science

Politicization: Space, Science

ANES: Federal spending on science and space vcf9048

13. Taxation

Politicization: Taxation

ANES: How Much Does the Federal Government Waste Tax Money vcf0606

14. Welfare & Poverty

Politicization: Welfare & Poverty

ANES: Federal spending on the poor vcf0886; Federal spending on homelessness vcf0893; Federal spending on welfare vcf0894; Federal spending on food stamps vcf9046

15. Workers

Politicization: Workers

ANES: Should government guarantee jobs vcf0809

Appendix J: Robustness to Term Frequency

A potential concern with the measurement of politicization scores in this paper is their relationship with term frequency: that is, whether changes in politicization reflect changes in how frequently terms in the dictionaries are used over time. To address this concern, I regressed each term-year’s politicization score on its frequency — the number of context windows contributing to that estimate — and used the residual, re-centered on the sample mean, as a frequency-adjusted politicization score. This removes any linear association between how many context windows entered a term-year’s estimate and its politicization score.

The overall trend under this adjustment is very similar in shape to the original (Figure J1). Figures J2 and J3 replicate Figures 4 and 5 using the frequency-adjusted score. The most- and least-politicized topics are also largely unchanged: race and immigration, religion, legal institutions, and gender and family remain among the most politicized issues, and home, outer space, and financial markets remain among the least. The main difference is that at the level of individual domains, the estimated increase in politicization is smaller on average once frequency is accounted for, and this is especially true for issues like education and foreign affairs.

Overall, this does not change the paper’s central conclusions: the broad rise in politicization since the 1990s and the ranking of the most and least politicized domains hold up even when frequency is controlled for, though the magnitude of the increase is somewhat sensitive to term frequency for a subset of domains. Importantly, I would caution against using the adjusted score as the “true” politicization score, because a term’s salience and its politicization are often substantively linked: a term can appear more frequently precisely because it has entered political contestation.

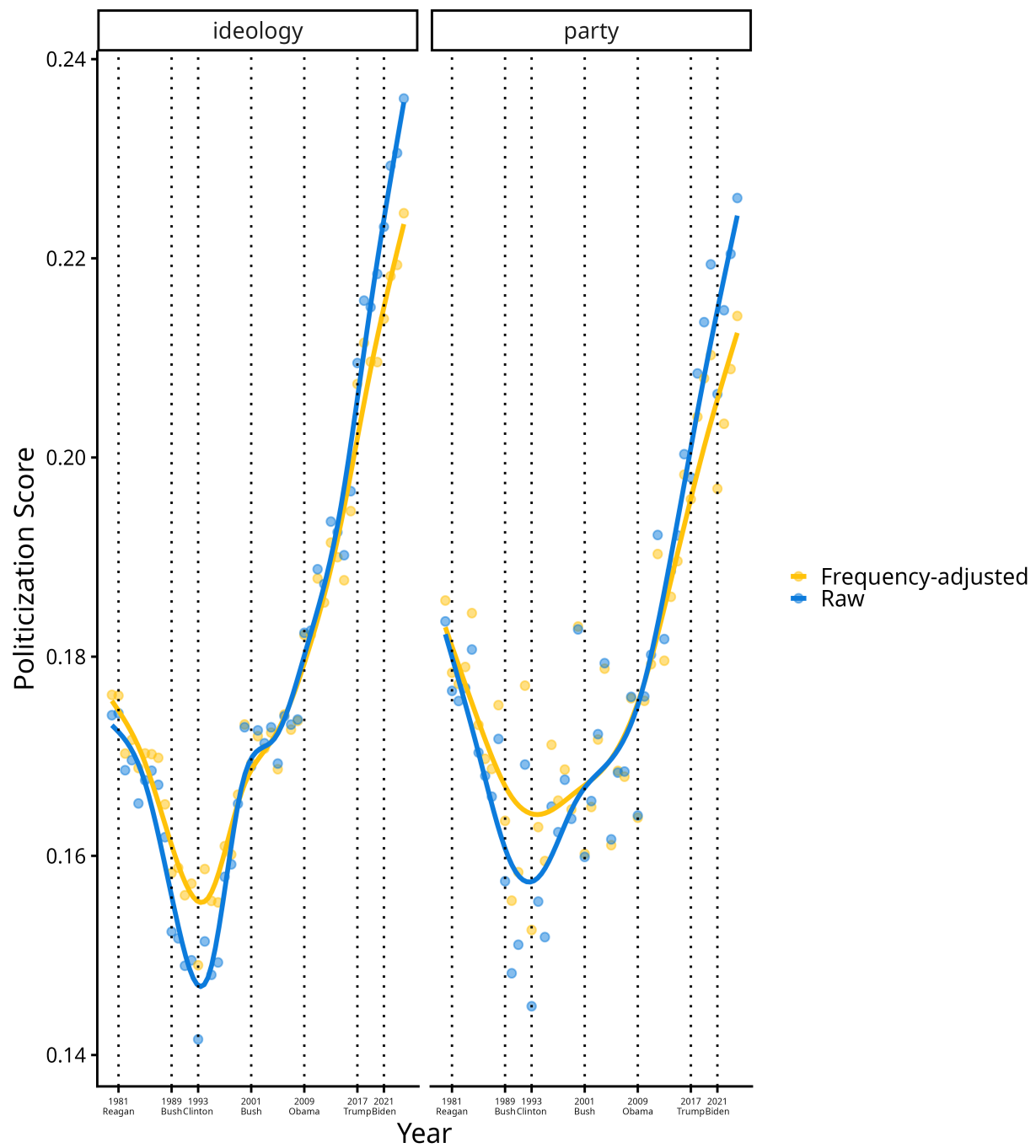


Figure J1. Average Ideological and Partisan Politicization, 1980–2024; Adjusted for Term Frequency vs. Original

Note: Trend lines are smoothed with a generalized additive model.

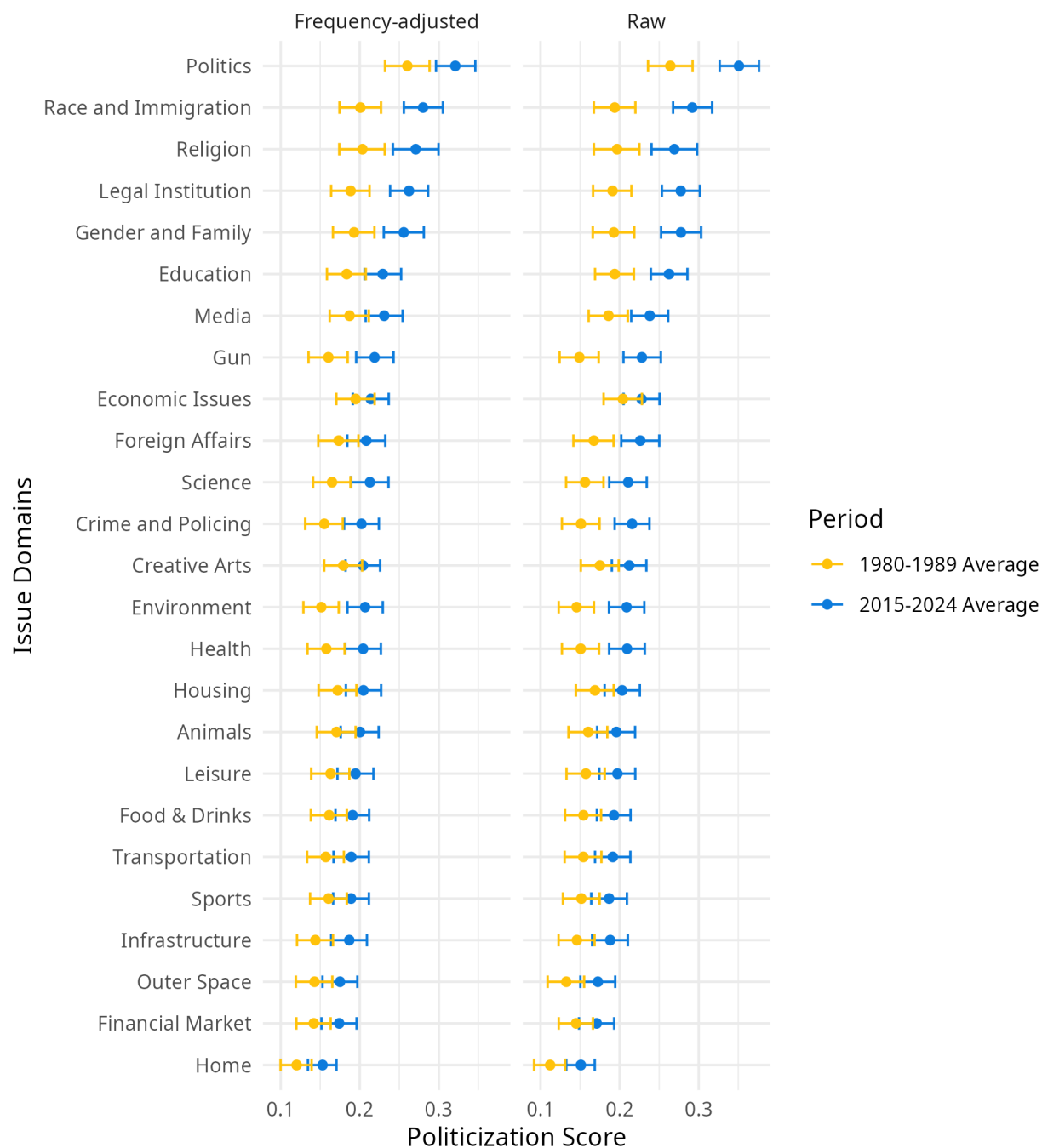


Figure J2. Levels of Ideological Politicization by Issue Domains, 1980-1989 Average and 2015-2024 Average; Adjusted for Term Frequency vs. Original

Note: Error bars around each point estimate show the 90% confidence interval, calculated by averaging the upper and lower bounds of the 90% confidence interval of the issue-year politicization scores within each 10-year period. Issues are sorted by their levels of politicization in 2015-2024.

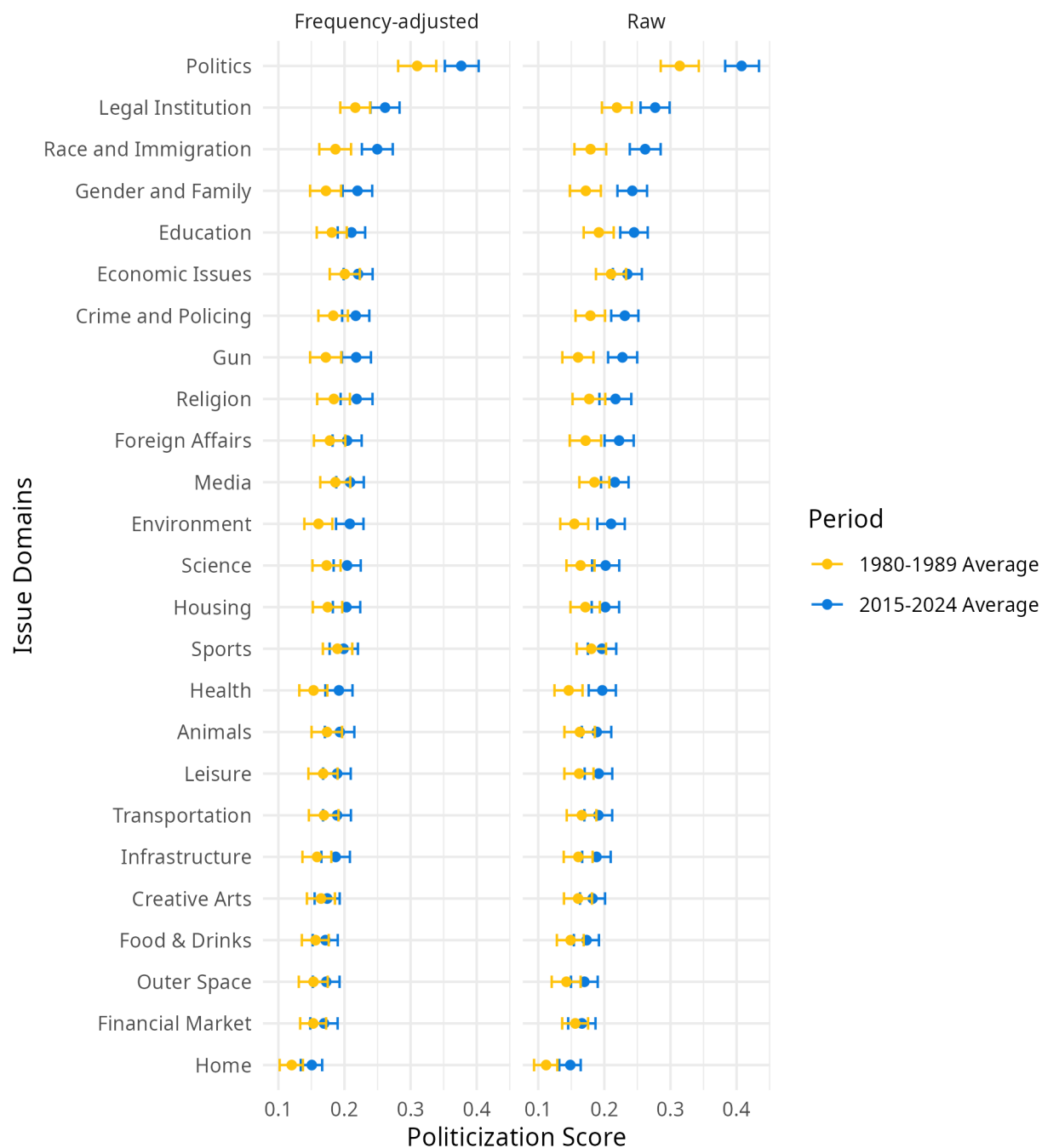


Figure J3. Levels of Partisan Politicization by Issue Domains, 1980-1989 Average and 2015-2024 Average; Adjusted for Term Frequency vs. Original

Note: Error bars around each point estimate show the 90% confidence interval, calculated by averaging the upper and lower bounds of the 90% confidence interval of the issue-year politicization scores within each 10-year period. Issues are sorted by their levels of politicization in 2015-2024.